

Solar Support Handbook

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Purpose and scope

This handbook gives support agents approved information for the fictional SolarFlow Home Controller SF-100. It covers product behavior, installation boundaries, error E14, warranty, maintenance, and safety escalation.

Support boundary

Use this handbook to explain documented steps. Do not invent missing product features, warranty terms, or repair instructions. Escalate when the handbook says to contact an authorized technician.

Product facts

- Product: SolarFlow Home Controller SF-100.
- Supported array: up to eight rooftop solar panels.
- Green LED: normal operation.
- Amber LED: the controller is waiting or completing a startup check.
- Red LED: the controller detected a fault that needs attention.
- The serial number and installation date appear on the label beside the lower cable entry.

Before giving advice

- Ask for the model name, LED color, displayed error code, and when the issue began.
- Ask whether a storm, power outage, installation change, or Wi-Fi password change occurred recently.
- Record the serial number, installation date, and actions already attempted.

Installation and operating conditions

Approved placement and inspection limits

Placement requirements

- Install the SF-100 indoors on a dry, stable wall.
- Keep at least 30 cm of clear space around the controller for airflow and inspection.
- Keep the AC isolator visible and accessible.
- Do not place the controller near dripping water, direct spray, fuel, or high heat.
- Only an authorized installer may complete fixed wiring or change internal connections.

Customer-safe checks

Allowed	Not allowed
Read the LED and error code	Open the controller casing
Use the external AC isolator	Touch exposed conductors
Inspect visible external connectors	Repair or replace internal parts
Record photos from a safe distance	Bypass a protection device

Installation escalation

If the site does not meet these conditions, stop the procedure and refer the customer to an authorized installer.

Error E14 recovery

AC input interruption

Error E14 means the controller detected an interruption in AC input. A recent power outage, isolator operation, or external connector issue may cause the alert.

Approved recovery procedure

1. Confirm that the displayed code is E14 and record the LED color.
2. Switch off the external AC isolator.
3. Wait 60 seconds. Do not open the controller casing during the wait.
4. Inspect only visible external connectors from a safe position. Do not touch damaged or exposed wiring.
5. If no damage is visible, restore power using the AC isolator.
6. Observe the controller during startup and record whether E14 returns.

Repeat fault rule

If E14 returns twice within 24 hours, leave the system off and contact an authorized technician.

After a storm

Do not touch wet equipment, fallen cables, or visibly damaged connectors. If it is safe to use the isolator, leave the system off and contact an authorized technician. Storm damage may affect warranty coverage.

Warranty and service records

Coverage terms for the SF-100 system

Coverage

- The SF-100 controller warranty lasts five years from the installation date.
- Solar panels have 12-year manufacturing-defect coverage from the installation date.
- A warranty claim requires the product serial number and installation date.
- Keep the installer invoice or commissioning record as proof of installation.

Exclusions

- Storm, flood, fire, and other external damage are excluded.
- Unauthorized modification, internal repair, or opening the controller casing is excluded.
- Damage caused by installation that does not follow the placement requirements is excluded.
- Cosmetic marks that do not affect operation are excluded.

Do not promise approval

Support agents may explain the documented coverage, but an authorized service review determines the final claim outcome.

Record for every service case

- Customer question and observed symptoms.
- Model, serial number, installation date, LED color, and error code.
- Recent outage, weather event, installation change, or app change.
- Steps completed and the result of each step.

Routine maintenance

Low-risk care and status checks

Monthly visual check

- Confirm the controller is dry and the ventilation space remains clear.
- Check the LED color and record any error code.
- Look for damage to visible external cables and connectors without touching them.
- Remove loose dust from the outside with a dry, soft cloth while following site safety rules.

Cleaning restrictions

- Do not spray water or cleaner on the controller.
- Do not use abrasive pads, solvents, or compressed air.
- Do not insert tools or cloth through ventilation openings.

When to schedule service

- A red LED remains on after an approved recovery procedure.
- The same fault returns repeatedly.
- External cables, connectors, or the wall mount appear damaged.
- The casing is unusually hot, swollen, cracked, or discolored.

Maintenance record

Record the inspection date, LED state, observed condition, and any service action. These notes help a technician reproduce intermittent faults.

Safety and escalation

Stop conditions and support response

Immediate stop conditions

- Smoke, flame, a burning smell, sparking, or unusual noise.
- A hot, swollen, cracked, wet, or discolored controller casing.
- Exposed conductors, fallen cables, or visible storm damage.
- Any condition in which reaching the AC isolator would be unsafe.

Response

- Keep people away from the equipment.
- Switch off the external AC isolator only when it is safe to do so.
- For smoke, flame, or immediate danger, contact local emergency services.
- Contact an authorized technician and provide the recorded case details.
- Do not restore power until the site has been declared safe.

Unknown information

If a customer asks about a feature or procedure that is absent from this handbook and current service alerts, say that the information could not be found in the supplied documents. Do not guess.

Escalation information

- Model and serial number.
- Installation date and installer record.
- LED color, error code, and exact sequence of events.
- Weather or power events, visible damage, and steps already completed.